Breaking the Loop: How Radia Perlman Made the Modern Internet Possible

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Process Paper

When asked to name one influential figure in technology, many will give one of few answers. Bill Gates, Steve Jobs, and Mark Zuckerberg are a few common examples. People who know more about the field may name figures such as Sam Altman, Jensen Huang, or Tim Berners-Lee. However, the name of the unsung hero who enabled the success of all the figures listed above is not well known to the public. Even within the Computer Science and Networking industries, their name is often unrecognized. However, within these industries, her work is revered, and easily recognized by almost all networking professionals. Her name is Radia Perlman, creator of the Spanning Tree Protocol (STP). Her work allowed for redundancy in early networks without causing catastrophic crashes, eliminating the tradeoff between redundancy and stability, laying the groundwork for the modern internet.

My research process began with reading general articles about Radia Perlman, and her work on STP. As I continued my research, the articles started to include more complex networking terms. Because of this, I started looking into the details of ARPANET, packet switching, broadcast storms, switching loops, and other networking nomenclature. After gaining an understanding of STP, I started researching primary sources pertaining to my topic, including Perlman's original paper. After my research was complete, I organized an interview with Chris Grabosky of MongoDB, to further research the use of STP in modern networking.

I built a website to highlight the importance of Radia Perlman and her work, as I felt it best represented her work in the Computer Science and Networking fields. The website format allows me to take full control of my submission's styling, and allows the experience I crafted to be led by the user. I started by mapping the path I would provide for the user, and then wrote paragraphs synthesizing my evidence for every step of the journey. I then selected media to assist and support my writing. I coded my website using Python’s Streamlit library, and supported it with HTML and CSS for additional styling. This allowed me to make my website more interactive, with buttons and dynamic elements.

Radia Perlman’s work with STP was not simply the creation of a protocol. Perlman actively solved a critical challenge preventing computer networks from growing. Her work transformed Ethernet from an unreliable system at volume to an easily scalable network, laying the foundations for the modern internet. This is my theme; how Perlman’s STP solved network scalability enabled the creation and growth of the modern internet. Her work did not just fix a technical bottleneck - STP made the modern internet possible at scale.